

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2016-17

Botany
Paper: BOB-101 — Cryptogams

Time: Three Hours

Full Marks: 70

Note: Attempt five questions in all, selecting two questions from each Section (A and B). Question No. 9 (Section C) is compulsory. The figures in the right-hand margin indicate marks.

SECTION – A

Question 1. Describe the characteristic features of Rhodophyceae. Give an illustrated account of reproduction in *Polysiphonia*. [15]

Question 2. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Sexual reproduction in *Sargassum*
- (b) Morphology and reproduction of *Nostoc*
- (c) Asexual reproduction in *Draparnaldiopsis*

Question 3. With the help of labeled diagrams, give a brief account of morphology and sexual reproductive structures of *Anthoceros*. [15]

Question 4. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Sexual reproduction in *Marchantia*
- (b) Cap cell and zoospore formation in *Oedogonium*
- (c) Sporangia of *Ectocarpus*

SECTION – B

Question 5. Give an illustrated account of various types of spore formation in *Puccinia*. [15]

Question 6. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Reproduction in *Albugo*
- (b) General characteristics of Ascomycotina
- (c) Morphology of gill and basidiospore formation in *Agaricus*

Question 7. With the help of suitable diagrams, give a brief account of morphology and reproduction in *Selaginella*. [15]

Question 8. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) General characteristics of Lycophyta
- (b) *Alternaria*
- (c) Reproduction in *Pteris*

SECTION – C

Question 9. Answer the following (in not more than 50 words each): [1 × 10 = 10]

- (a) Name the reserve food material present in the Xanthophyceae.
- (b) What are heterokont flagella?
- (c) Name the pigments present in Cyanophyceae.
- (d) Zygosporangium is formed in which fungus?
- (e) Name the fungal group which possesses Crozier.
- (f) What is the dikaryotic phase?
- (g) What is the function of the receptacle in *Sargassum*?
- (h) What is the function of the Vallecular canal?
- (i) What are the xerophytic characteristics of *Equisetum* stem?
- (j) What is the difference between an elater and a pseudoelater?

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